higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N6

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This marking guideline consists of 7 pages.

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QUESTION 1

1.1 $F_f = \mu NR$
$= 0,12(1\ 200 + 180)$
$= 165,6\ N$ (2)

1.2 $\theta = 60° \times \frac{\pi}{180°} = 1,047\ rad$
$= (1,047)(0,03)$
$= 31,42\ mm$
$WD = F \times S$ (¿ *overcome friction*)
$= (165,5)(31,42 \times 10^{-3})$
$= 5,2\ J$
$WD = FS$ (¿ *move the follower*)
$= (1\ 380)(0,01)$
$= 13,8\ J$
$WD_{TOTAL} = 13,8 + 5,2$
$= 19\ J$
$T_q = \frac{WD_{TOTAL}}{\theta} = \frac{19}{1,047} = 18,147\ Nm$ (11)

1.3 *Resultant force acting on cam*:
$F_R = \sqrt{F_f^{\,2} + N_R^{\,2}}$
¿ $\sqrt{(165,6)^2 + (1\ 380)^2}$
$= 1\ 389,9\ N$

Bending moment ∈ the shaft:
$M_{MAX} = W\,l$
$= (1\ 389,9)(0,2)$
$= 277,98\ Nm$ (4)

1.4 $T_e = \sqrt{T_{MAX}^{\,2} + M_{MAX}^{\,2}}$
$T_e = \sqrt{(18,147)^2 + (277,98)^2}$
$T_e = 278,572\ Nm$
$T_e = \frac{\pi}{16} D^3 \tau_{MAX}$
$278,572 = \frac{\pi}{16} D^3 (45 \times 10^6)$
$D = 31,59\ mm$
$\therefore D = 32\ mm$
$M_e = \frac{1}{2}[M_{MAX} + T_e]$
¿ $\frac{1}{2}[277,98 + 278,572]$
$= 278,272\ Nm$

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$$M_e = \frac{\pi}{32} D^3 \sigma_n$$

$$278{,}276 = \frac{\pi}{32} D^3 \left(60 \times 10^6\right)$$

$$D = 36{,}15 \ \ mm$$

$$\therefore D = 37 \ \ mm$$

$$\therefore Safe \ \ diameter \ \ D = 37 \ \ mm$$ (9)

1.5

$$\frac{T_1}{T_2} = e^{\mu\theta} \quad \frac{T_1}{T_2} = e^{(0{,}45)(2{,}793)}$$

$$T_1 = 3{,}514\, T_2$$

$$T_q = \left(T_1 - T_2\right) R$$

$$18{,}147 = \left(3{,}514\, T_2 - T_2\right) 0{,}4$$

$$T_2 = 17{,}854 \ \ N$$

$$T_1 = 62{,}74 \ \ N$$ (5)

1.6

$$M_{MAX} = 277{,}98$$

$$W_P = mg = 40 \times 9{,}81 = 392{,}4 \ \ N$$

$$M_{MAX} = \left(W_p + T_1 + T_2\right) L$$

$$277{,}98 = \left(392{,}4 + 62{,}74 + 17{,}854\right) L$$

$$L = 588 \ \ mm$$ (5)

1.7

$$F_V = 392{,}4 + 62{,}74 + 17{,}854 + 1\ \ 380 = 1\ \ 852{,}994 \ \ N$$

$$F_H = 165{,}6 \ \ N$$

$$F_R = \sqrt{F_H{}^2 + F_V{}^2}$$

$$F_R = \sqrt{(165{,}6)^2 + (1\ \ 852{,}994)^2}$$

$$F_R = 1\ \ 860{,}379 \ \ N$$ (4)

1.8

$$F_T = \mu F_R$$

$$= (0{,}08)(1\ \ 860{,}379)$$

$$= 148{,}83 \ \ N$$

$$P = \frac{F_T}{A}$$

$$39\ \ 166 = \frac{148{,}83}{A}$$

$$A = 3{,}8 \times 10^{-3} \ \ m^2$$

$$A = DL$$

$$3{,}8 \times 1\,0^{-3} = (0{,}037)\, L$$

$$L = 100 \ \ mm$$ (6)

1.9 *Frictional torque ∧ power*:

$$T_f = F_T \times R_{SHAFT}$$
$$T_f = (148,83)(18,5 \times 10^{-3})$$
$$T_f = 2,753 \ \ Nm$$
$$P_f = \frac{2\pi N T_f}{60}$$

$$P_f = \frac{2\pi(1\ 000)(2,753)}{60}$$
$$P_f = 288,331 \ \ W$$
$$Heat_g = Heat_L$$
$$\dot{m}\, c\, \Delta t = Power_L$$
$$\dot{m}(3,187)(8) = 288,331$$
$$\dot{m} = 0,0113 \ \ kg/s$$
$$Q = \frac{\dot{m}}{\delta_{REL}}$$
$$0,064 = \frac{0,0113}{\delta_{REL}}$$
$$\delta_{REL} = 0,177$$

(8)
[54]

QUESTION 2

2.1 2.1.1

$$C = \frac{m}{6}(T_A + T_B)$$
$$285 = \frac{m}{2}(31 + 64)$$
$$m = 6 \ \ mm$$
$$C = \frac{m}{2}(T_C + T_D)$$
$$285 = \frac{6}{2}(T_C + 53)$$
$$T_C = 42 \ \ teeth$$
$$C = \frac{m}{2}(T_E + T_F)$$
$$285 = \frac{6}{2}(37 + T_F)$$
$$T_F = 58 \ \ teeth$$
$$C = \frac{m}{2}(T_G + T_H)$$
$$285 = \frac{6}{2}(T_G + 67)$$

$T_G = 28 \ \ teeth$

$$\frac{N_H}{N_A} = \frac{T_A \times T_G}{T_B \times T_H}$$

$$\frac{N_H}{1\ \ 500} = \frac{31 \times 28}{64 \times 67}$$

$N_H = 303{,}738 \ \ r/min$

$$\frac{N_F}{N_A} = \frac{T_A \times T_E}{T_F \times T_B}$$

$$\frac{N_F}{1\ \ 500} = \frac{31 \times 37}{58 \times 64}$$

$N_F = 463{,}497 \ \ r/min$

$$\frac{N_D}{N_A} = \frac{T_A \times T_C}{T_B \times T_D}$$

$$\frac{N_D}{1\ \ 500} = \frac{31 \times 42}{64 \times 53}$$

$N_D = 575{,}767 \ \ r/min$

$N_{OUT} = 1\ \ 500 \ \ r/min$ (15)

2.1.2

$$m = \frac{PCD_G}{T_G}$$

$$6 = \frac{PCD_G}{28}$$

$PCD_G = 168 \ \ mm$

$$m = \frac{PCD_H}{T_H}$$

$$6 = \frac{PCD_H}{67}$$

$PCD_H = 402 \ \ mm$

$$R_{MAX} = \sqrt{R^2 + 2Rr\sin^2\varphi + r^2\sin^2\varphi}$$

$$¿ \sqrt{(201)^2 + 2(201)(84)\sin^2 20 + (84)^2\sin^2 20}$$

$= 212{,}547 \ \ mm$

$$R_{ACTUAL} = \frac{outside \ \ diameter}{2}$$

$$= \frac{402 + 2 \times 6}{2}$$

$= 207 \ \ mm$

$No \ \ interference \ \ R_{MAX} > R_{ACTUAL}$ (7)

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2.1.3

$$m=\frac{PCD_F}{T_F}$$
$$6=\frac{PCD_F}{58}$$
$$PCD_F=348\ \ mm$$
$$V=\frac{\pi ND}{60}$$
$$¿\frac{\pi(463{,}497)(0{,}348)}{60}$$
$$=8{,}445\ \ m/s$$

Velocity factor:
$$C_V=\frac{3}{3+V}$$
$$C_V=\frac{3}{3+8{,}445}$$
$$C_V=0{,}262$$

Working stress:
$$\theta=\theta_o\times C_V$$
$$=(140\times10^6)(0{,}262)$$
$$=36{,}68\ \ MPa$$

Tangental force:
$$F_T=\frac{P}{V}$$
$$¿\frac{55\times10^3}{8{,}445}$$
$$=6{,}51\ \ kN$$

Form factor:
$$y=0{,}154-(\frac{0{,}912}{T})$$
$$=0{,}154-\frac{0{,}912}{37}$$
$$=0{,}129$$

Face width:
$$F_T=\pi\times m\times b\times y\times\theta$$
$$6\ \ 513=\pi(0{,}006)b(0{,}129)(36{,}68\times10^6)$$
$$b=73\ \ mm$$
$$b_{min}=2{,}5\times\pi\times m$$
$$=2{,}5\times\pi\times6$$
$$=47{,}12$$
$$b_{max}=4\times\pi\times m$$
$$=4\times\pi\times6$$
$$=75{,}4$$

(15)

2.2 *Pressure*:

$$P=\frac{F_n}{A}$$

$$200\times10^3=\frac{F_n}{\pi\left(R^2-r^2\right)}$$

$$F_n=628\ \ 319\left(R^2-r^2\right)N$$

Power:

$$p=\frac{2\pi NT}{60}$$

$$55\times10^3=\frac{2\pi\left(1\ \ 500\right)T}{60}$$

$$T=350{,}141\ \ Nm$$

$$T=\mu\times F_n\times R_m$$

$$350{,}141=0{,}45\left(628\ \ 319\right)\left(R^2-r^2\right)\left[\frac{2}{3}\left(\frac{R^3-r^3}{R^2-r^2}\right)\right]$$

$$1{,}858\times10^{-3}=R^3-r^3$$

$$1{,}858\times10^{-3}=R^3-\left(\frac{2}{3}R\right)^3$$

$$R^3=2{,}64\times10^{-3}$$

$$R=138{,}214\approx139\ \ mm$$

$$r=\frac{2}{3}R=\frac{2}{3}\left(139\right)=93\ \ mm$$

(9)
[46]

TOTAL: 100

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